

EG7030 Dissertation

Supervisor Weekly Update — Week 5

Student: Usha Rani Vamanagiri (u2965962)
Programme: MSc Electric Vehicles and Energy Storage
Date: 17 July 2026
Module: EG7030 — Dissertation

1. Research Question

This dissertation proposes, validates, and evaluates the **Fleet Stress Index (FSI)** — a composite, physics-grounded feature index designed to bridge the gap between laboratory battery datasets (constant-current cycling, $KI = 0$) and real-world EV fleet operation (highly variable current profiles, $KI = 0.60\text{--}0.81$). The central question is whether FSI can improve SoH prediction accuracy and provide early degradation warning for commercial EV fleets.

2. Work Completed This Week

FSI Validation Pipeline — All 5 Datasets Complete

Dataset	Chemistry	Key Result	Status
CALCE (UMD)	LiCoO ₂	RMSE 3.73%, $R^2 = 0.9839$ (XGBoost)	Training
NASA PCoE RW	LiCoO ₂	Cross-profile validation passed	Validated
Oxford LFP	LFP	Cross-chemistry validation passed	Validated
NREL BLAST-Lite	NMC/LFP/NCA	RMSE 20.4%; 84% is calibration bias	Validated
Severson 2019	LFP	Spearman(FSI, -cycle life) $\rho = 0.81$, $p < 0.001$	Validated

Fleet DNA KI Validation

Vehicle Class	Trips	KI Mean	KI Std
Delivery Trucks	553	0.604	0.076
Transit Buses	472	0.631	0.084
Refuse Trucks	387	0.813	0.118
CC Lab Baseline	—	0.000	0.000

Wilcoxon signed-rank tests confirm $KI \gg 0$ for all vehicle classes ($p < 0.001$). This validates the core premise: lab CC cycling ($KI = 0$) cannot replicate the stress profile of real fleet operation.

Project Restructure and Documentation

- Restructured project into 7 numbered academic reproducibility folders (Raw → Code → Processed → Results → Dissertation)
- Created **DATA_DICTIONARY.csv** — complete column definitions, FSI formula derivation, and dataset provenance for all 5 datasets
- Submitted 1,500-word progress report (weekly update form, 17 July 2026)

3. Key Findings

- FSI correctly ranks degradation severity** across 23 distinct fast-charging protocols (Severson LFP): Spearman $\rho = 0.81$, $p < 0.001$. Cells on highest-KI protocols degrade 62%

faster than single-step protocols.

- **KI = 0 in all lab CC data; KI = 0.60–0.81 in all real fleet data** — confirming the training-deployment gap that FSI is designed to close.
- **Cross-chemistry performance gap is calibration, not structure.** 84% of the 20.4% BLAST RMSE is a chemistry-specific offset removable by one-parameter linear rescaling. SHAP feature rank correlation CALCE vs BLAST: $\rho > 0.80$.
- **XGBoost in-sample performance (CALCE):** RMSE = 3.73%, $R^2 = 0.9839$. SHAP confirms KI as the dominant feature by importance margin.

4. Question for Supervisor Guidance

T_{norm} Asymmetry — Scope or Fix?

The current thermal stress term is $T_{\text{norm}} = |T_{\text{avg}} - 25| / 25$, which is *symmetric* — it treats 10°C and 40°C identically (both give $T_{\text{norm}} = 0.60$). This contradicts the Arrhenius degradation model, where elevated temperature accelerates degradation more than cold of the same magnitude from reference.

An asymmetric alternative would be:

- $T > 25^\circ\text{C}$: $T_{\text{norm}} = (T - 25) / 25$ (positive, Arrhenius-aligned)
- $T \leq 25^\circ\text{C}$: $T_{\text{norm}} = (25 - T) / 50$ (half-weight for cold, since cold slows reactions rather than accelerating degradation)

Should I implement this correction within the dissertation scope, or state it as a clearly-scoped limitation and recommend it as future work? I am aware it would require re-running the full validation pipeline and recalibrating FSI weights.

5. Plan for Next Week

- Write **Chapter 3 — Methodology**: FSI formula derivation, dataset extraction pipeline, ML training protocol
- Write **Chapter 4 — Results**: All five dataset validation results, cross-validation matrix, SHAP analysis
- Generate **publication-quality figures**: KI bar chart by vehicle class, FSI vs SoH scatter, SHAP beeswarm, protocol ranking chart
- Begin **Chapter 5 — Discussion**: calibration gap framing, T_{norm} limitation (pending supervisor guidance above), fleet deployment implications

All code, data, and validation results are version-controlled. The full 1,500-word progress report and DATA_DICTIONARY are available on request.